



PulseNetwork

Real-Time Blood Donation & Smart Inventory Analytics

A high-end responsive Web Application for Bangladesh divisions

<https://pulsenetwork.onrender.com>

- Traditional lists: Phone books and Facebook pages lead to outdated contact numbers.
- Geographic separation: Patients cannot visualize or find closest available donors easily.
- Verification issues: Lack of live verification of blood bank stocks leads to time wastage.
- No Demand Forecast: Hospitals face unexpected shortages without predictive analytics.

An AI-driven live ecosystem connecting patients, donors, and blood banks instantly.

Interactive Division Map

100% geographically correct 8-division map of Bangladesh with glowing live status nodes.

Instant Live Chat

Secure real-time chat between patient, donor, and hospitals via WebSockets.

Smart Stock Analytics

Verified live inventory controls with ML algorithms predicting blood demand trends.

- A real-time heartbeat SVG wave illustrating live activity scores and counts.
- Donors who log donation are automatically locked for 90 days to protect health.
- Perfect rendering on laptops, tablets, and phones with stacked headers and spacing.
- 1-click deployment with persistent storage (render.yaml specification).

Frontend Layer

HTML5, Vanilla CSS3 (Neon layout), JavaScript (Single Page Routing).

Backend Server

Node.js, Express (RESTful APIs, session authentications).

Real-Time Sync

ws WebSocket Server (Instant chat dispatch, broadcast alerts).

Predictive Analytics

Embedded Machine Learning calculations for regional forecasts.

Storage Engine

JSON persistent data engine configured with Redis persistent disks.

PulseNetwork — Saving Lives in Real-Time

Live Site: <https://pulsenetwork.onrender.com>

GitHub: <https://github.com/CoderRinku/PulseNetwork>